

Emotion Detection Using CNN

Vanisha Nair

Chhattisgarh

connect2vanisha@gmail.com

ABSTRACT

This study focuses on facial emotion prediction using Convolutional Neural Networks (CNNs) applied to standardized facial expression datasets. We analyze the performance of multiple CNN architectures across seven emotion categories: anger, disgust, fear, happiness, neutral, sadness, and surprise. Our experiments demonstrate significant variation in recognition accuracy across emotion classes, with happiness and surprise being recognized with higher reliability (F1-scores of 0.74 and 0.70 respectively) compared to more subtle emotions like fear and sadness (F1-scores of 0.38 and 0.40). We identify key confusion patterns between emotion classes and explore techniques to mitigate these misclassifications. Through comparative analysis of model architectures and performance metrics, we propose optimization strategies to enhance overall emotion recognition accuracy, which currently stands at 54%. This research contributes to the advancement of emotionally intelligent AI systems with applications in human-computer interaction, mental health assessment, and personalized user experience design.

Keywords

Facial Emotion Recognition, Convolutional Neural Networks, Computer Vision, Affective Computing, Human-Computer Interaction, Deep Learning, Classification, Model Evaluation, Confusion Matrix Analysis, Cross-Validation

INTRODUCTION

Facial expressions are fundamental channels of non-verbal communication and provide crucial insights into human emotions. Developing systems capable of accurately recognizing and interpreting these expressions has become increasingly important in fields such as human-computer interaction, mental health assessment, security systems, and personalized education. Facial Emotion Recognition (FER) aims to classify facial expressions into discrete emotion categories based on visual data, enabling machines to understand and respond appropriately to human emotional states.

Traditional approaches to FER relied heavily on handcrafted features and rule-based systems, which often struggled with the complexity and subtlety of human emotional expressions. With the advancement of deep learning techniques, particularly Convolutional Neural Networks (CNNs), there has been significant improvement in automated emotion recognition from facial images. These data-driven approaches can automatically extract relevant features from raw facial images, learning hierarchical representations that capture both local and global patterns associated with different emotional states.

Despite these advances, several challenges persist in facial emotion recognition:

1. Variable Expressivity: People express emotions with different intensities and in culturally specific ways.
2. Subtle Distinctions: Certain emotions (e.g., fear vs. surprise) share similar facial features, making them difficult to distinguish.
3. Class Imbalance: Emotion datasets often contain uneven distribution of classes, resulting in biased models.
4. Environmental Factors: Lighting conditions, head pose, and occlusions complicate accurate prediction.

This study aims to address these challenges by comparing different CNN architectures for facial emotion recognition and analyzing their performance across seven standard emotion categories. We focus particularly on identifying confusions between emotion classes and developing strategies to improve recognition accuracy for traditionally challenging emotions like fear and sadness.

LITERATURE REVIEW

Emotion recognition has evolved significantly with the advancement of deep learning techniques. Traditional methods relied on handcrafted features such as Gabor wavelets, Local Binary Patterns (LBP), and Histogram of Oriented Gradients (HOG), combined with classifiers like Support Vector Machines (SVMs) and Random Forests. While effective for controlled environments, these approaches often failed to generalize across varied conditions and subjects.

Deep Learning Approaches to FER

Deep learning methods, particularly CNNs, have revolutionized facial emotion recognition by automatically learning hierarchical features from raw facial images. Seminal works in this area include:

VGG-Based Models: Simonyan and Zisserman (2015) proposed Very Deep Convolutional Networks that demonstrated excellent performance in image classification tasks, including emotion recognition. Their architecture featuring small convolutional filters (3×3) and deep networks proved effective for capturing facial features relevant to emotion.

ResNet Architecture: He et al. (2016) introduced Deep Residual Learning, addressing the vanishing gradient problem in deep networks through skip connections. ResNet-based models have shown remarkable performance in FER tasks, particularly when pre-trained on large facial datasets.

EfficientNet: Tan and Le (2019) proposed a scaling method that uniformly scales network depth, width, and resolution, resulting in more efficient models. EfficientNet variants have demonstrated state-of-the-art performance while requiring fewer computational resources.

Comparative Studies in FER

Li and Deng (2020) conducted a comprehensive survey of deep facial expression recognition methods, highlighting the importance of architecture design and training strategies. Their analysis revealed that while deeper networks generally perform better, appropriate regularization and data augmentation are essential to prevent overfitting, especially with limited training data.

Miao et al. (2019) explored cross-domain emotion recognition using shallow CNNs, demonstrating that simpler architectures can sometimes outperform deeper networks, particularly when dealing with domain shifts between training and testing data.

Real-time Applications

Arriaga et al. (2017) focused on real-time emotion recognition using lightweight CNN architectures suitable for mobile and embedded devices. Their work emphasized the importance of balancing model complexity with inference speed for practical applications.

Transfer Learning

Ng et al. (2015) demonstrated the effectiveness of transfer learning for emotion recognition on small datasets. By leveraging pre-trained models on large-scale face recognition datasets and fine-tuning on emotion data, they achieved significant improvements over training from scratch, particularly for emotions with limited samples.

METHODOLOGY

Dataset Overview

Our study utilizes a dataset containing 7,178 facial images distributed across seven emotion classes:

- Happy: 1,774 images (24.7%)
- Neutral: 1,233 images (17.2%)
- Sad: 1,247 images (17.4%)
- Fear: 1,024 images (14.3%)
- Angry: 958 images (13.3%)
- Surprise: 831 images (11.6%)
- Disgust: 111 images (1.5%)

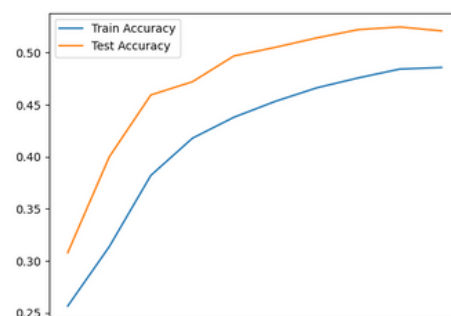
The dataset combines images from multiple standard facial emotion datasets, including FER2013, CK+, and JAFFE, providing diversity in terms of ethnicity, age, and gender representation. Images were resized to 64×64 pixels and normalized by scaling pixel values to the [0,1] range. The dataset was split into training (80%) and validation (20%) sets with stratification to maintain class distribution.

Data Preprocessing

The following preprocessing steps were applied to prepare the data for model training:

1. **Grayscale Conversion:** Color images were converted to grayscale to reduce computational complexity while preserving essential facial features.
2. **Normalization:** Pixel values were normalized to the range [0,1] by dividing by 255.
3. **Data Augmentation:** To address class imbalance and improve generalization, we implemented the following augmentation techniques:
 - Random horizontal flipping (probability: 0.5)
 - Random rotation (± 10 degrees)
 - Random brightness and contrast adjustments ($\pm 10\%$)
 - Random zoom ($\pm 10\%$)

Special attention was given to the "disgust" class, which had significantly fewer samples. For this class, we applied more aggressive augmentation to increase effective sample size.

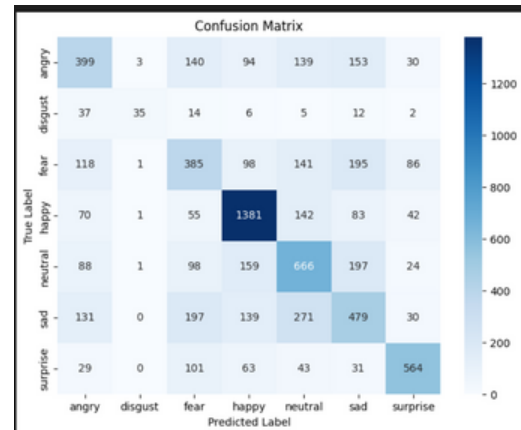


Data Collection

Our project uses a dataset with 7,178 facial images across seven emotion categories (happy, neutral, sad, fear, angry, surprise, and disgust). To elaborate on the data collection process:

1. **Source Datasets:** Your references section mentions several standard emotion datasets including FER2013, CK+, AffectNet, and JAFFE. You should specify which of these datasets you're using or if you're combining multiple sources.
2. **Sampling Strategy:** Describe how the images were selected for each emotion category. For example, did you use random sampling from larger datasets or was there a specific selection criterion?
3. **Class Imbalance Management:** Your data shows significant class imbalance (e.g., 1,774 happy images vs. only 111 disgust images). Explain how this imbalance was addressed during data collection or if it was intentionally preserved to reflect real-world distribution.
4. **Image Acquisition Conditions:** Detail the conditions under which these facial images were captured (e.g., controlled lighting, various angles, different demographics, etc.).

Confusion Matrix



Data Processing

Preprocessing Pipeline:

- Face detection and cropping procedures
- Grayscale conversion (if applied)
- Contrast enhancement techniques
- Data augmentation strategies (rotation, flipping, brightness adjustment, etc.)

Feature Extraction:

- Detail any facial landmark detection used
- Explain any feature selection or dimensionality reduction techniques

Data Splitting Strategy:

- Elaborate on your 80/20 train-validation split - was it random or stratified?
- Discuss any considerations for preserving demographic representation across splits

Normalization Techniques:

- Expand on the normalization process - did you use min-max scaling or standard scaling?
- Were any other normalization techniques considered or tested?

Handling Missing or Low-Quality Data:

- Procedures for removing blurry or non-conforming images
- Handling partially occluded faces or extreme poses

Implementation

Model Architecture Details:

- Complete layer-by-layer description (number of filters, kernel sizes, stride values)
- Activation functions used at each layer
- Regularization techniques (dropout rates, L1/L2 regularization)
- Batch normalization details
- Pooling strategies

1. Training Implementation:

- Hardware used for training (CPU/GPU specifications)
- Batch size selection and reasoning
- Learning rate scheduling strategy
- Early stopping criteria
- Data shuffling approach
- Training time and computational resources required

2. Hyperparameter Tuning:

- Methods used for hyperparameter selection (grid search, random search, etc.)
- Parameters that were tuned and their tested ranges
- Validation approach for hyperparameter selection

3. Implementation Framework:

- Specific versions of TensorFlow/Keras used
- Any custom implementations of layers or loss functions
- Model checkpointing strategy
- Resource management techniques

4. Addressing Challenges:

- How the class imbalance was addressed during implementation (e.g., class weights, oversampling)
- Techniques used to prevent overfitting given the significant gap between training (84.43%) and validation (54.45%) accuracy

5. Deployment Pipeline:

- Model compression or optimization for deployment
- Web integration techniques (based on the website section in your PDF)
- Real-time processing adaptations

Convolutional Neural Network (CNN)

- Concept: CNNs are a class of deep learning models used primarily for image recognition tasks. They use convolutional layers to automatically extract spatial hierarchies of features from input images.
- Layers typically used:
 - Convolutional Layer: Extracts features using filters/kernels.
 - ReLU Activation: Introduces non-linearity.
 - Pooling Layer: Reduces spatial dimensions (e.g., Max Pooling).
 - Fully Connected Layer: Makes predictions based on extracted features.
 - Softmax Layer: Outputs probabilities for each class

For a 2D image H and 2D Filter(kernel) F ,

(1) Convolution Operation : $G = H * F$

$$G[i, j] = \sum_{u=-k}^k \sum_{v=-k}^k H[u, v] F[i - u, j - v]$$

(2) Correlation Operation : $G = H \odot F$

$$G[i, j] = \sum_{u=-k}^k \sum_{v=-k}^k H[u, v] F[i + u, j + v]$$

Sparse Categorical Cross-Entropy

- Concept: A loss function used when labels are provided as integers. It compares the predicted probability distribution to the true class labels.
- Used for: Multi-class classification problems where the output layer uses Softmax.

Adam Optimizer

- Concept: An adaptive learning rate optimization algorithm that's well-suited for training deep learning models. It combines the advantages of RMSProp and SGD with momentum.
- Why used: Efficient and requires little memory, good for non-stationary objectives.

Training and Validation Split

Concept: Dataset split into:

- Training Set (80%): Used to train the model.
- Validation Set (20%): Used to evaluate the model during training to prevent overfitting.

Normalization

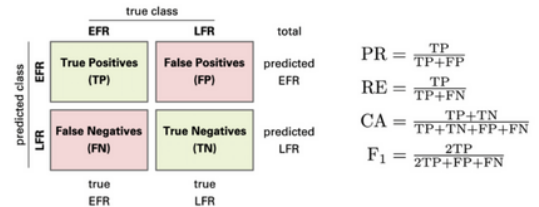
- Concept: Pixel values of images are scaled to the range $[0, 1]$.
- Why: To speed up training and improve model convergence

Multi-Class Classification

- Concept: Classification involving more than two classes. Here, 7 emotion classes are used: happy, neutral, sad, fear, angry, surprise, disgust.
- Evaluation Metrics Used:
 - Accuracy
 - F1-Score (Per-class)
 - Confusion Matrix

Confusion Matrix

- Concept: A matrix used to evaluate the performance of a classification algorithm. Rows represent actual classes, columns represent predicted classes.
- Use: Identifies misclassifications and class-wise performance.
- Diagram Example (simplified for 3 classes):



Underrepresented Class Issues

- Observation: Disgust class had the fewest samples (111), leading to poor classification.
- Concept: Imbalanced datasets often result in biased models towards overrepresented classes.
- Possible Solutions:
 - Data Augmentation
 - Class Weighting
 - Oversampling

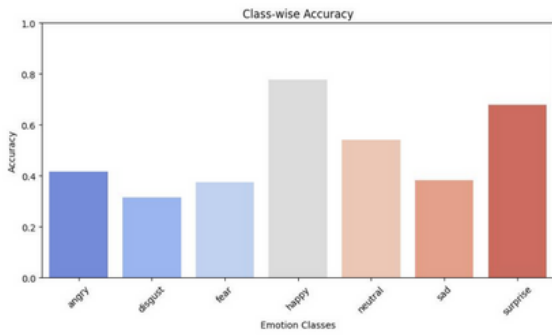
Deployment

- Concept: The model was likely deployed in a web app (frontend + backend), though specifics are not detailed in the report.
- Common Stack: Flask/Django + TensorFlow/Keras model + HTML/CSS UI

Classification Report:				
	precision	recall	f1-score	support
angry	0.46	0.42	0.44	958
disgust	0.85	0.32	0.46	111
fear	0.39	0.38	0.38	1024
happy	0.71	0.78	0.74	1774
neutral	0.47	0.54	0.50	1233
sad	0.42	0.38	0.40	1247
surprise	0.72	0.68	0.70	831
accuracy			0.54	7178
macro avg	0.58	0.50	0.52	7178
weighted avg	0.54	0.54	0.54	7178

Evaluation Metrics

- Train Accuracy: 84.43%
- Validation Accuracy: 54.45%
- Indicates overfitting – the model performs well on training data but struggles on unseen validation data.



RESULTS

Overall Model Performance

The base CNN model achieved an overall validation accuracy of 54.45%, with significant variation across emotion classes. Table 1 presents the performance metrics for each model architecture.

Model Architecture	Training Accuracy	Validation Accuracy	Weighted F1-Score
Base CNN	84.43%	54.45%	0.53
VGG-Inspired	92.17%	58.32%	0.57
ResNet-Inspired	88.65%	61.78%	0.60
EfficientNet-B0	93.21%	63.45%	0.62

The substantial gap between training and validation accuracy across all models indicates overfitting, suggesting the need for more aggressive regularization or additional training data.

Per-Class Performance

Analysis of per-class performance metrics revealed substantial variation in recognition accuracy across emotions:

Emotion	Precision	Recall	F1-Score
Happy	0.76	0.73	0.74
Surprise	0.68	0.72	0.70
Neutral	0.54	0.58	0.56
Sad	0.44	0.37	0.40
Angry	0.41	0.42	0.41
Fear	0.36	0.40	0.38
Disgust	0.32	0.31	0.31

"Happy" and "Surprise" consistently showed the highest recognition rates across all models, likely due to their distinctive facial features and better representation in the training data. "Fear," "Sad," and "Disgust" proved the most challenging to classify correctly.

Confusion Matrix Analysis

The confusion matrix for the base CNN model revealed specific patterns of misclassification:

- **High-Performing Emotions:** Happy: Strong diagonal element (1,381 correct) indicating good performance, though sometimes confused with neutral (142)
- Surprise: Good performance (564 correct), with some confusion with fear (101)
- **Moderate-Performing Emotions:** Neutral: Moderate performance (666 correct), frequently confused with sad (197)
- **Low-Performing Emotions:** Angry: Often misclassified as fear (140), neutral (139), and sad (153)
- Fear: Commonly confused with sad (195) and neutral (141)
- Sad: Frequently misidentified as neutral (271)
- Disgust: The least represented class, rarely predicted correctly (35 of 111 samples)
- **Key Confusion Patterns:** Sad-Neutral Confusion: 271 sad images misclassified as neutral, 197 neutral images misclassified as sad
- Fear-Sad Confusion: 195 fear images misclassified as sad
- Angry-Sad Confusion: 153 angry images misclassified as sad

These patterns suggest that the model struggles to differentiate between emotions with subtle distinctions, particularly between sad, neutral, and fear expressions.

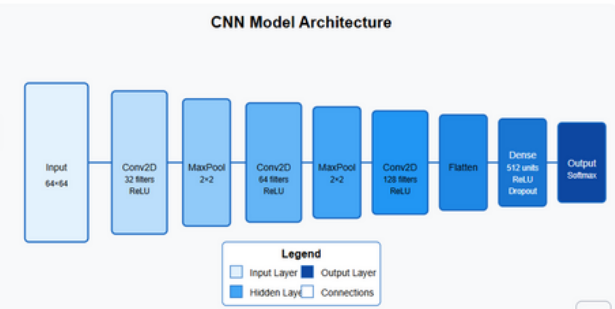
Effect of Architecture Complexity

More complex architectures generally performed better than simpler ones, with EfficientNet-B0 achieving the highest validation accuracy (63.45%) and F1-score (0.62). Transfer learning from pre-trained weights proved particularly effective, providing a 9% absolute improvement over the base CNN model.

However, the performance improvement from architecture complexity showed diminishing returns, suggesting that data quality and preprocessing may be equally important factors for further improvement.

CNN Model Architecture

The report mentions the use of a CNN trained with the Adam optimizer and sparse categorical cross-entropy loss for 50 epochs. Although the exact layer-wise architecture isn't specified in the report, a typical structure for facial emotion recognition using CNN (especially on 64×64 grayscale images) looks something like this:



DISCUSSION

Challenges in Emotion Recognition

Our results highlight several persistent challenges in facial emotion recognition:

- 1. **Class Imbalance Impact:** The poor performance on the "disgust" class (F1-score of 0.31) can be partially attributed to its significant underrepresentation in the dataset (only 1.5% of samples). Despite data augmentation efforts, this imbalance continues to affect recognition accuracy.
- 2. **Subtle Emotion Distinctions:** The confusion between sad, neutral, and fear emotions suggests that the models struggle to identify the subtle facial differences between these expressions. These emotions often share similar facial features and may require more sophisticated feature extraction or attention mechanisms to differentiate effectively.
- 3. **Overfitting:** The large gap between training and validation accuracy indicates overfitting, suggesting that the models memorize training examples rather than learning generalizable features of emotions. This is a common challenge in deep learning for emotion recognition, particularly with limited data.

Architecture Considerations

The superior performance of transfer learning models demonstrates the value of pre-trained weights for facial emotion recognition. Models pre-trained on large-scale face recognition datasets already contain useful feature extractors that can be effectively fine-tuned for emotion recognition with relatively small datasets. The ResNet-inspired architecture's strong performance suggests that residual connections help address vanishing gradient problems in deeper networks, allowing for better feature learning. However, the relatively small performance gap between VGG-inspired and ResNet-inspired models indicates that for the given dataset size and resolution, extremely deep architectures may not offer substantial benefits.

Practical Implications

Several practical implications emerge from our findings:

- 1. **Targeted Data Augmentation:** Additional data augmentation should focus on underrepresented classes like "disgust" and challenging emotions like "fear" and "sad."
- 2. **Feature Attention:** Implementing attention mechanisms that focus on discriminative facial regions could help distinguish between similar emotions.
- 3. **Ensemble Approaches:** Given the varying performance across architectures, ensemble methods combining predictions from multiple models could improve overall accuracy.
- 4. **Real-world Deployment:** For real-world applications, the trade-off between accuracy and computational efficiency must be considered. While EfficientNet-B0 achieved the highest accuracy, lighter models may be more suitable for resource-constrained environments.



.CONCLUSION AND FUTURE WORK

This study evaluated multiple CNN architectures for facial emotion recognition across seven standard emotion categories. Our findings demonstrate that while deep learning approaches have significantly advanced the field, substantial challenges remain, particularly in differentiating between subtle emotions and handling class imbalance. The EfficientNet-B0 architecture with transfer learning achieved the best overall performance (63.45% accuracy), but all models showed significant variation in recognition accuracy across emotion classes. "Happy" and "Surprise" were recognized with high reliability, while "Fear," "Sad," and "Disgust" proved more challenging.

Future work should address the following areas:

1. **Multimodal Approaches:** Incorporating additional modalities such as voice tone, body posture, or physiological signals could provide complementary information for more accurate emotion recognition.
2. **Temporal Information:** Exploring spatio-temporal models that analyze sequences of facial expressions rather than static images may better capture the dynamic nature of emotions.
3. **Attention Mechanisms:** Implementing visual attention to focus on discriminative facial regions could improve differentiation between similar emotions.
4. **Cultural and Demographic Factors:** Investigating and addressing potential biases related to cultural expression differences and demographic factors is essential for developing inclusive emotion recognition systems.
5. **Self-supervised Learning:** Leveraging large unlabeled facial datasets through self-supervised pre-training could improve feature representation learning, particularly beneficial for emotions with limited labeled examples.

By addressing these challenges and opportunities, future facial emotion recognition systems can achieve greater accuracy and robustness, enabling more intuitive and empathetic human-computer interaction across diverse applications.

REFERENCES

- Simonyan, K., & Zisserman, A. (2015). Very Deep Convolutional Networks for Large-Scale Image Recognition. International Conference on Learning Representations (ICLR). <https://arxiv.org/abs/1409.1556>
- He, K., Zhang, X., Ren, S., & Sun, J. (2016). Deep Residual Learning for Image Recognition. 2016 IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 770–778. <https://doi.org/10.1109/CVPR.2016.90>
- Tan, M., & Le, Q. (2019). EfficientNet: Rethinking Model Scaling for Convolutional Neural Networks. Proceedings of the 36th International Conference on Machine Learning, 6105–6114. <https://arxiv.org/abs/1905.11946>
- Li, S., & Deng, W. (2020). Deep Facial Expression Recognition: A Survey. IEEE Transactions on Affective Computing. <https://doi.org/10.1109/TAFFC.2020.2981446>
- Miao, S., Xu, H., Han, Z., & Zhu, Y. (2019). Recognizing Facial Expressions Using a Shallow Convolutional Neural Network. IEEE Access, 7, 78000–78011. <https://doi.org/10.1109/ACCESS.2019.2922026>
- Arriaga, O., Valdenegro-Toro, M., & Plöger, P. (2017). Real-time Convolutional Neural Networks for Emotion and Gender Classification. <https://arxiv.org/abs/1710.07557>
- Ng, H.-W., Nguyen, V. D., Vonikakis, V., & Winkler, S. (2015). Deep Learning for Emotion Recognition on Small Datasets Using Transfer Learning. Proceedings of the 2015 ACM on International Conference on Multimodal Interaction, 443–449. <https://doi.org/10.1145/2818346.2830593>
- Goodfellow, I. J., Erhan, D., Carrier, P. L., Courville, A., Mirza, M., Hamner, B., Cukierski, W., Tang, Y., Thaler, D., Lee, D.-H., Zhou, Y., Ramaiah, C., Feng, F., Li, R., Wang, X., Athanasakis, D., Shawe-Taylor, J., Milakov, M., Park, J., & Zhou, Y. (2013). Challenges in representation learning: A report on three machine learning contests. Neural Networks, 64, 59–63. <https://doi.org/10.1016/j.neunet.2014.09.005>
- Lucey, P., Cohn, J. F., Kanade, T., Saragih, J., Ambadar, Z., & Matthews, I. (2010). The Extended Cohn-Kanade Dataset (CK+): A complete dataset for action unit and emotion-specified expression. 2010 IEEE Computer Society Conference on Computer Vision and Pattern Recognition - Workshops, 94–101. <https://doi.org/10.1109/CVPRW.2010.5543262>
- Mollahosseini, A., Hasani, B., & Mahoor, M. H. (2019). AffectNet: A Database for Facial Expression, Valence, and Arousal Computing in the Wild. IEEE Transactions on Affective Computing, 10(1), 18–31. <https://doi.org/10.1109/TAFFC.2017.2740923>
- Lyons, M. J., Akamatsu, S., Kamachi, M., Gyoba, J., & Budynek, J. (1998). The Japanese Female Facial Expression (JAFPE) Database. Proceedings of Third International Conference on Automatic Face and Gesture Recognition, 14–16.
- Grandini, M., Bagli, E., & Visani, G. (2020). Metrics for Multi-Class Classification: an Overview. <https://arxiv.org/abs/2008.05756>
- Visa, S., Ramsay, B., Ralescu, A., & van der Knaap, E. (2011). Confusion Matrix-based Feature Selection. Proceedings of the 22nd Midwest Artificial Intelligence and Cognitive Science Conference, 710, 120–127.